

DR. GUOCHENG QIAN

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PROFESSIONAL EXPERIENCE

Snap Inc.

- Research Scientist
- Topic: Generative Models, Personalization

Palo Alto, CA, USA

May 2024 - Present

Snap Middle East Fz (Snap Inc. Dubai office)

- Research Scientist
- Topic: 3D Deep Learning, Generative Models

Dubai, UAE

Dec 2023 - April 2024

EDUCATION

King Abdullah University of Science and Technology (KAUST)

Ph.D. in Computer Science, specializing in 3D Vision

Jeddah, Saudi Arabia

May 2019 - Dec 2023

Xi'an Jiaotong University (XJTU)

Bachelor's degree in Engineering with honors
Outstanding Undergraduate

Xi'an, China

Aug 2014 - Jul 2018

INTERNSHIP EXPERIENCE

Snap Inc.

- Research Scientist Intern, working with Sergey Tulyakov
- paper: "AToM: Amortized Text-to-Mesh using 2D Diffusion" (under review)
- paper: "Magic123: One image to High-Quality 3D Object Generation using diffusion priors" (ICLR'24)

May 2023 - Sep 2023

Meta Platforms

- AI Research Scientist Intern, working with Yunyang Xiong, Haoqi Fan, Zhuang Liu

Aug 2022 - Jan 2023

SenseTime Research

- Computer Vision Intern, mentored by senior director Dr. Jimmy S. Ren and Dr. Dong Chao
- Paper: "Rethinking Learning-based Demosaicing, Denoising, and Super-Resolution Pipeline" (ICCP'22)

Aug 2018 - Apr 2019

PUBLICATIONS

See google scholar for a full list: <https://scholar.google.com/citations?user=DUDaxg4AAAAJ>

Citations: 1821; H-index: 14; First-authored papers: 10; Top-tier conference/journal papers: 17. (Date: 12/04/2024)

- [1] **Guocheng Qian**, Jinjie Mai, Abdullah Hamdi, Jian Ren, Aliaksandr Siarohin, Bing Li, Hsin-Ying Lee, Ivan Skorokhodov, Peter Wonka, Sergey Tulyakov, and Bernard Ghanem. **Magic123: One image to High-Quality 3D Object Generation** using diffusion priors. In *International Conference on Learning Representations (ICLR)*, 2024
- [2] **Guocheng Qian**, Xingdi Zhang, Abdullah Hamdi, and Bernard Ghanem. Pix4Point: Image pretrained transformers for 3d point cloud understanding. In *International Conference on 3D Vision (3DV)*, 2024
- [3] **Guocheng Qian**, Yuchen Li, Houwen Peng, Jinjie Mai, Hasan Hammoud, Mohamed Elhoseiny, and Bernard Ghanem. Pointnext: Revisiting pointnet++ with improved training and scaling strategies. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 23192–23204. Curran Associates, Inc., 2022
- [4] **Guocheng Qian***, G. Li*, Matthias Müller*, Itzel C. Delgadillo, Abdullellah Abualshour, Ali K. Thabet, and Bernard Ghanem. **DeepGCNs: Making gcns go as deep as cnns**. *IEEE transactions on pattern analysis and machine intelligence (T-PAMI)*, 45(6):6923–6939, 2023

- [5] **Guocheng Qian**, Hasan Hammoud, Guohao Li, Ali Thabet, and Bernard Ghanem. Assanet: An anisotropic separable set abstraction for efficient point cloud representation learning. In *Advances in Neural Information Processing Systems (NeurIPS Spotlight)*, volume 34, pages 28119–28130. Curran Associates, Inc., 2021
- [6] **Guocheng Qian**, Abdullellah Abualshour, Guohao Li, Ali Thabet, and Bernard Ghanem. PU-GCN: Point cloud upsampling using graph convolutional networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 11683–11692, June 2021
- [7] **Guocheng Qian***, G. Li*, Itzel C. Delgadillo*, Matthias Müller, Ali K. Thabet, and Bernard Ghanem. SGAS: Sequential greedy architecture search. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1617–1627, 2020
- [8] **Guocheng Qian**, Yuanhao Wang, Jinjin Gu, Chao Dong, Wolfgang Heidrich, Bernard Ghanem, and Jimmy S Ren. Rethinking learning-based demosaicing, denoising, and super-resolution pipeline. In *IEEE International Conference on Computational Photography (ICCP)*, pages 1–12. IEEE, 2022
- [9] **Guocheng Qian**, Xuanyang Zhang, Guohao Li, Chen Zhao, Yukang Chen, Xiangyu Zhang, Bernard Ghanem, and Jian Sun. When NAS meets trees: An efficient algorithm for neural architecture search. In *CVPR Workshop*, pages 2782–2787, 2022
- [10] Yash Kant, Ziyi Wu, Michael Vasilkovsky, **Guocheng Qian**, Jian Ren, Riza Alp Guler, Bernard Ghanem, Sergey Tulyakov, Igor Gilitschenski, and Aliaksandr Siarohin. Spad: Spatially aware multiview diffusers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024
- [11] Abdullah Hamdi, Luke Melas-Kyriazi, **Guocheng Qian**, Jinjie Mai, Ruoshi Liu, Carl Vondrick, Bernard Ghanem, and Andrea Vedaldi. Ges: Generalized exponential splatting for efficient radiance field rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024
- [12] Chen Zhao, Shuming Liu, Karttikeya Mangalam, **Guocheng Qian**, Fatimah Zohra, Abdulmohsen Alghannam, Jitendra Malik, and Bernard Ghanem. Dr²net: Dynamic reversible dual-residual networks for memory-efficient finetuning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024
- [13] Jun Chen, Deyao Zhu, **Guocheng Qian**, Bernard Ghanem, Zhicheng Yan, Chenchen Zhu, Fanyi Xiao, Mohamed Elhoseiny, and Sean Chang Culatana. Exploring open-vocabulary semantic segmentation without human labels. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023
- [14] Sicheng Chen, **Guocheng Qian**, Bernard Ghanem, Yongqing Wang, Zhou Shu, Xuefeng Zhao, Lei Yang, Xinqin Liao, and Yuanjin Zheng. Quantitative and real-time evaluation of human respiration signals with a shape-conformal wireless sensing system. *Advanced Science*, 9(32):2203460, 2022
- [15] Tianyu Zhang, **Guocheng Qian**, Jin Xie, and Jian Yang. Fastpci: Motion-structure guided fast point cloud frame interpolation. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024
- [16] Jinjie Mai, Wenxuan Zhu, Sara Rojas Martinez, Jesus Zarzar, Abdullah Hamdi, **Guocheng Qian**, Bing Li, Silvio Giancola, and Bernard Ghanem. Tracknerf: Bundle adjusting nerf from sparse and noisy views via feature tracks. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024

ACADEMIC EXPERIENCE

Teaching Assistant Deep Learning for Visual Computing (2020 - 2022), Design and Analysis of Algorithms (2021)

Reviewer CVPR, ECCV, ICCV, ICML, NeurIPS, AAAI, IJCAI, T-PAMI, TVCG, SIGGRAPH Asia

Conference Organizer Tutorial on Graph Machine Learning for Computer Vision @ CVPR'22 , Workshop on Learning 3D with Multi-View Supervision @ CVPR'23&CVPR'24

HONORS & AWARDS

KAUST CEMSE Research Excellence Award Less than 10 students

KAUST CEMSE Dean's List Award Awarded to 20% CEMSE students

KAUST Fellowship Full tuition support, monthly living allowance, housing, and medical coverage

National Scholarship Top 1%, first-class national scholarship in China

Outstanding Undergraduate Highest undergraduate honor awarded to 10 selected undergraduates

Excellent Student Cadre Awarded to 3% undergraduate students with leadership

PROGRAMMING SKILLS

- **PyTorch**: Magic123 🌟>1,500 stars, PointNeXt 🌟>500 stars, SGAS 🌟>150 stars, DeepGCNs 🌟>1000 stars
- **Large scale GPU usage**: multi-node distributed training experience using over 100 GPUs
- **C++, CUDA**: Course projects on **CUDA** based image processing (🌟), and **C++** based curvature estimation (🌟)
- **MATLAB**: experience in image processing, **3D reconstruction** (🌟)